

Bradley Lignoski

Math Specialist, Full Stack Engineer & 4.0 AI Graduate Student

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Technologies and Skills

- Python, Pytorch, Scikit Learn, Numpy, Pandas, Seaborn, Matplotlib,
- ML Pipelines and Custom Transformers
- Dimensionality Reduction, Feature Engineering/Ranking/Selection, Model Tuning
- Containerization & APIs: Docker, Flask, Node, Terraform
- Automated Testing, CI/CD
- Semantic Embeddings, RAG
- Cloud Deployment: Azure Machine Learning, Terraform, Linux (15+ years)
- Model Evaluation & Drift Detection
- Agile, Jira
- SQL, Migrations
- Git, Github, Azure Repositories
- Linux user for over 15 years
- Technical writing/communicating
- Extensive mathematics and physics background

Experience

AI Systems Engineer (Graduate Project), Johns Hopkins University Sep 2024 – Dec 2024

Built containerized APIs and supporting AI systems for fraud detection, facial recognition, worksite monitoring, and personalized movie recommendations, focusing on scalable ML pipelines, real-time inference, extensive evaluation and tuning and automated monitoring.

- Built containerized real-time fraud detection API, leveraging a modular ML pipeline that supports dynamic model and threshold selection, ensuring robust classification.
- Designed and built an adaptive movie recommendation engine, implementing collaborative filtering, content-based filtering and cold-start solutions, enhancing user engagement through nDCG@k-optimized reranking.
- Developed a FAISS-powered facial recognition system, enabling low-latency nearest-neighbor search across millions of embeddings using VGGFace2 and CASIA-WebFace models.
- Optimized object detection for worksite environments, integrating YOLO-based inference, non-maximum suppression (NMS), and hard negative mining, refining predictions and dramatically improving precision and recall scores.
- Built an augmentation pipeline simulating real-world noise (brightness, blur) and synthetic ground-truth bounding box transformations to evaluate model robustness.
- Implemented statistical drift detection (KS test on RMSE) to detect performance degradation.
- Containerized AI models using Flask & Docker, ensuring reproducible deployment and structured logging for real-time error tracking.

Full-Stack Software Engineer, Convoy Inc – Seattle, WA Sept 2021 - Feb 2023

- Served on the Contract Team at Convoy, developing crucial tools to streamline the negotiation of freight contracts for over 400,000 clients
- Operated in a fast-paced Agile environment, contributing across the stack to three different React applications and multiple Apollo-NodeJS microservices
- Designed and built system for validating and recording location data to flexibly validate location clusters for high value customers
- Performed requirements gathering and implemented feature flag configuration, built RPC endpoints and corresponding client code
- Implemented an advanced hotkey feature within AG-Grid, reducing time account managers spent on certain tasks by 99%. In-depth requirement gathering, research, and testing
- Strategically restructured hotkey system for future extensibility
- Wrote and monitored time-expensive batched SQL backfill affecting 11 million mission-critical records, unblocking the data science tech debt initiative
- Proactively mitigated technical debt and enhanced software reliability by identifying and addressing gaps in existing unit tests during resolution of over 200 different software bugs

Professional Development Projects – Seattle, WA May 2023 - Dec 2023

- Advanced Technical Training: Developed [an easily extensible, fully containerized Node.js server](#) integrated with CircleCI, utilizing Docker to ensure CircleCI and local integration test compatibility
- Enrolled in Masters of Science, AI & Machine Learning, Johns Hopkins Whiting School of Engineering

App Developer, Freelance – Seattle, WA Oct 2017 - Sept 2019

Founder and full-stack developer of a [mobile-hybrid social fitness app for gym climbers](#) to create, share, and discover user-generated climbing routes.

- Built an interactive route-editing map interface using a custom `svg-pan-zoom` super-class, supporting image-based route annotation and intuitive gesture controls
- Developed a vector projection algorithm to map screen taps to real-world climbing wall positions, enabling spatial encoding, route search, and filtering.
- Designed an image cache prioritization system using RxJS and Cordova, optimizing bandwidth and storage for low-end mobile devices.
- Authored over 15 custom components and ~30 cloud event handlers (via Google Cloud Functions) to support commenting, gesture handling, and asynchronous database cleanup
- Implemented a dictionary-order spatial indexing algorithm to synchronize map movements with a live route preview carousel.
- Built automated consensus metrics using serverless functions to update community-sourced difficulty and popularity scores.

Previous Career: Curriculum Developer, Teacher, Math Department Chair

Northwest School - Seattle, WA

Hillside Student Community - Bellevue WA

- Taught and developed computer science curriculum with Replit, Python and Pytest
- Taught and developed advanced calculus, & linear algebra curriculum
- Led a team of 6 educators, improving subject cohesion, demonstrating effective leadership while fostering a collaborative environment
- Redesigned prospective math teacher vetting with tech-style interview component

Khan Academy - Mountain View CA (remote)

- Using JS and LaTeX, built 1,400 interactive online problem sets, currently used by about 3 million globally-distributed users
(example course <https://www.khanacademy.org/math/math3>)
- Improved quality and speed of the peer-review process by writing custom Python scripts to automate error checking, eliminating 100% of relevant errors for critical tasks

Education

Johns Hopkins Whiting School of Engineering Feb 2024 - present

Masters of Science in Artificial Intelligence and Machine Learning | 4.0 GPA

Expected completion: September 2025

- **ML Theory & Applications:** Bias-variance tradeoff, regularization (L1/L2, dropout), feature engineering, probabilistic modeling, and core ML paradigms (supervised, unsupervised, reinforcement learning, state space search and constraint satisfaction problems).
- **Deep Learning & Model Optimization:** Neural network fundamentals (backpropagation, activation functions), modern architectures (CNNs, RNNs, transformers), and training techniques.
- **Hyperparameter Optimization:** Grid & random search (baseline methods), Bayesian optimization (probabilistic search), and Hyperband (adaptive tuning for deep learning models).
- **Evaluation & Interpretability:** Model performance metrics (AUC-ROC, AUC-PRC, F1-score), loss functions (cross-entropy, contrastive, triplet), and explainability methods (SHAP, LIME, feature importance).
- **Mathematical Foundations:** Eigenvalues, SVD, PCA, Bayesian inference, Markov chains, Entropy, and calculus for ML (gradients, automatic differentiation).
- **Data Engineering & Representation:** Preprocessing (missingness, outliers, class imbalance, etc), NLP techniques (tokenization, embeddings), and graph-based ML (node classification, embeddings, knowledge graphs).

The College of Wooster, Wooster OH - *B.A. Major: Physics, Minor: Mathematics, Independent Study: Quantum Decoherence Theory*